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LESSON PLAN IN SCIENCE 10

Ms. Althea C. Calinawan **Sir John Steve C. Landiza** Teaching Intern	**January 27-30, 2026** Date of Teaching
Ma'am Blezy V. Guiroy Teaching Internship Mentor	**Grade 10** Grade Level
January 26, 2026 Date of Submission	**9:30 A.M-2:15 P.M** Time Schedule

Content Standard	*The learners demonstrate an understanding of:* - how gases behave based on the motion and relative distances between gas particles
Learning Competencies	*The learners shall be able to investigate the relationship between:* - volume and pressure at constant temperature of a gas; - volume and temperature at constant pressure of a gas; and - explain the relationships using Kinetic Molecular Theory.

I. Intended Learning Outcomes (Objectives)

At the end of the lesson and through varied learning activities, the Grade 10 learners with at least 60% proficiency will be able to:

a. analyze Boyle's and Charle's Laws using the Kinetic Molecular Theory;
b. demonstrate appreciation for the two laws and their daily uses;
c. manipulate a water bottle to physically verify the relationship between volume and pressure;
d. interpret the results of the balloon experiment to explain the effect of temperature on gas volume; and
e. solve problems related to Boyle's and Charles' Laws.

II. Learning Content

A. **Topic (s)**	- **Boyle's Law and Charle's Law**
B. **Key Concepts**	(1) Boyle's Law describes the relationship between pressure and volume of a gas. - It states that at constant temperature, the pressure of a gas is inversely proportional to its volume. If the volume decreases, the pressure increases, and vice versa, as long as temperature remains constant. (2) Charles's Law describes the relationship between volume and temperature of a gas. - It states that at constant pressure, the volume of a gas is directly proportional to its absolute temperature (Kelvin). - If the temperature increases, the volume increases, and if the temperature decreases, the volume decreases, as long as pressure is constant.
C. **References** *(Print, electronic, etc.)*	- SCIENCE 10 Learner's Module, Department of Education, Republic of the Philippines https://drive.google.com/file/d/13Hki4N89NxOUNh5so8Foyij3d1pUm1tE/view?usp=sharing https://drive.google.com/file/d/1jCY254BdYAwN0jPWVy3vx1HxmNeIFnv3/view - Libretexts. (2020, August 11). *6.3 Gas Laws - Boyle's and Charles' laws*. Chemistry LibreTexts. https://chem.libretexts.org/Courses/Los_Angeles_Trade_Technical_College/Hybrid_Chem_51_v1/07%3A_Gases/6.3_Gas_Laws_-_Boyle's_and_Charles'_Laws - Gas Properties - Ideal Gas Law | Kinetic Molecular Theory | Diffusion - PhET Interactive Simulations - Javalab. (2025, November 3). Charles's Law. https://javalab.org/en/charles_law_en/ - Şeyma Sucu. (2023, March 27). Gases are compressible #science #experiment #physics #marshmallow #chemistry [Video]. YouTube. https://www.youtube.com/watch?v=PsvvytZZZUs

	- Playing With Rain. (2021, February 2). Hot and Cold Balloon Experiment [Video]. YouTube. https://www.youtube.com/watch?v=tzjHwwFTaBI
D. **Instructional Materials**	- Laptop - Canva Presentation
E. **Integration** *(Values, SDG/ESD, GAD, etc.)*	- **Quality Education (SDG 4):** Enhances analytical thinking and scientific literacy by utilizing the Kinetic Molecular Theory to visualize invisible particle interactions, enabling students to logically predict and calculate physical changes in gas properties. - **Good Health and Well-being (SDG 3)**: Promotes physiological awareness and accident prevention by applying gas laws to vital functions like breathing mechanics and recognizing the dangers of pressurized containers (e.g., heated aerosols, scuba safety), fostering safer daily practices. - **Values Integration (Prudence, Situational Awareness, and Precision)**: Cultivates a culture of safety and critical observation by emphasizing the volatility of gases under changing conditions, reinforcing the need for caution and precision when managing temperature and pressure variables to prevent hazards.
F. **Skills** *(21st century skills)*	- Critical Thinking and Problem Solving - Creativity and Innovation - Collaboration
Preliminaries	- Prayer - Attendance Checking
III. Teaching and Learning Process (Experiences)	

I. Recapitulation (3 minutes)

The learners will recall key concepts about the properties of gases from the previous lesson. They will answer the following questions:

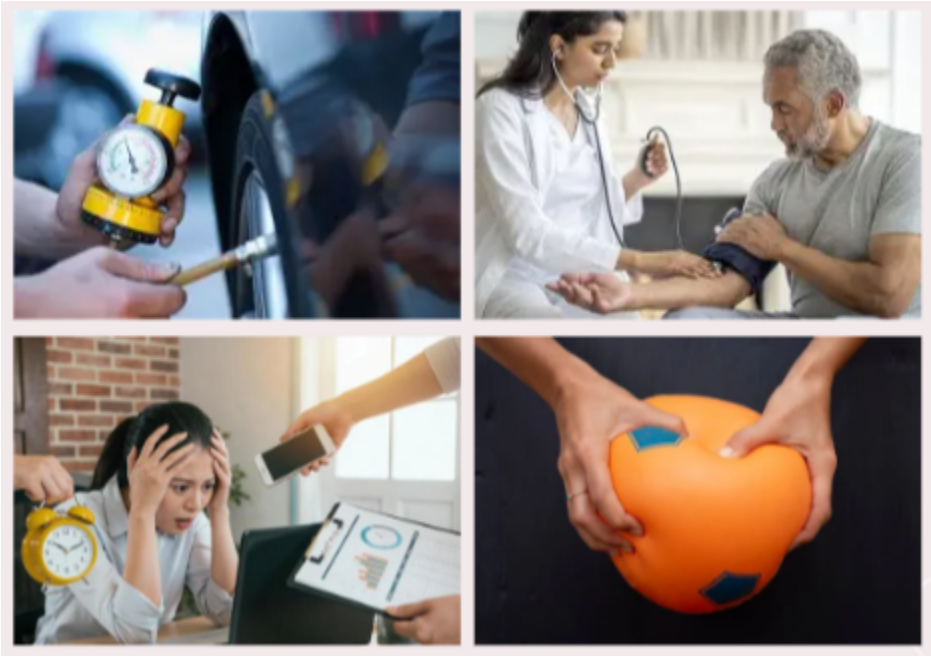
a. What are the four physical properties of gas?
   Pressure, temperature, volume, and amount of gas

b. It is the scientific model that explains the behavior of gasses.
   Kinetic Molecular Theory

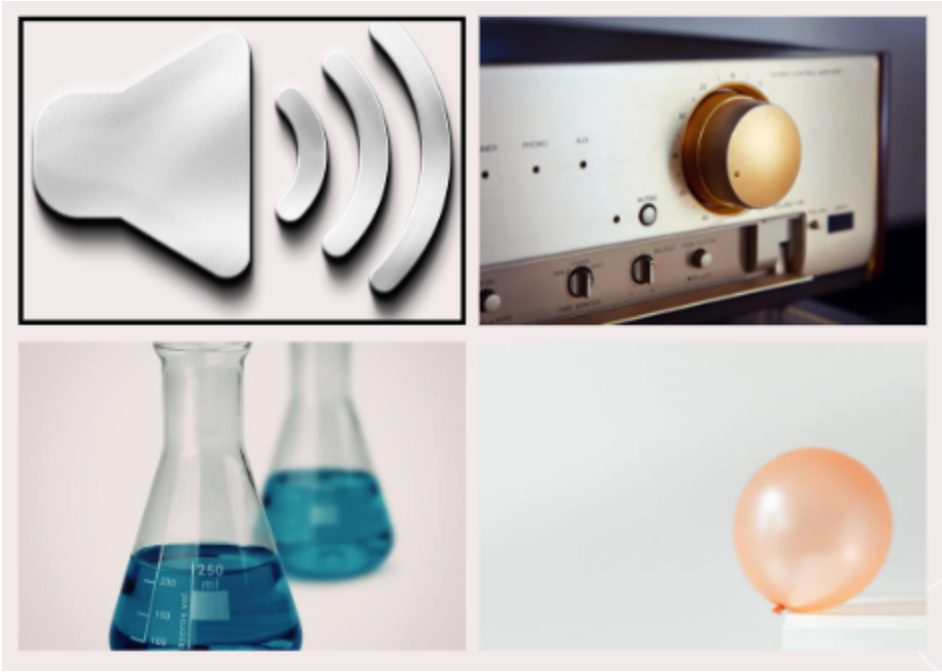
c. How do the gas particles behave when they are in a hot environment? Cold environment?
   When gas particles are in a hot environment, they move faster and collide more often with each other and the walls of the container. When gas particles are in a cold environment, they move slower and collide less often, taking up less space if the container can shrink.

II. Motivation (5 minutes)

The learners will be playing 4 pics one word depicting the following:



Pressure



Volume

Compress

After the quick game, the students will be asked what their idea for the topic of the day is.

A. Activity (5 minutes)

1. Boyle's Law

The students will be divided into 4 groups. Each group will have a plastic water bottle that is tightly sealed. The teacher will give the task of opening the water bottle by only using one twist of the cap. The groups will be given a 3 minute think time, and 2 minutes to present their solution.

If none of the groups are able to accomplish this task, the teacher will show it themselves.

- The trick is to squeeze the bottle tight first, then give the cap a quick twist.

Groups that accomplish the task are given 5 extra points

2. Charle's Law

The learners will watch a short video demonstrating the expansion and contraction of gas as temperature changes. They will observe the balloon and note how its volume changes with temperature. Link of the video is as follows:

https://youtu.be/tzjHwwFTaBI?si=sOhUUoScVEDYTg8g

Afterwards, students will observe a Charles's Law simulation that shows how the volume of gas changes with temperature at constant pressure.

https://javalab.org/en/charles_law_en/

B. Analysis (5 minutes)

1. Boyle's Law

After the demonstrations conclude, the teacher will ask the students guided questions to explain the solution, focusing on the concepts of pressure, volume, and their relationship.

Guide Questions:

1. Describe what happened to the space inside the bottle when you twisted the bottom. Did the space get bigger or smaller?
    - Smaller
2. When you twisted the bottle body, how did the plastic feel? Did it become softer or harder? Why do you think that happened?
    - Harder, particles are pushing against the cap at higher frequency
3. We didn't add any more air into the bottle. So, why did the air push the cap off with so much force?
    - Particles are pushing against the cap at higher frequency (higher pressure)
4. If the air particles are like tiny bouncing balls, what happened to their 'bouncing' when we forced them into a smaller space?
    - They have less room to move, and thus more collisions against the wall occur
5. Based on this activity, what is the relationship between the Volume (size of the space) and the Pressure (force of the air)? As one goes down, what happens to the other?

- Inversely proportional, goes higher

2. Charle's Law

After the demonstrations conclude, the learners will answer the following guided questions:

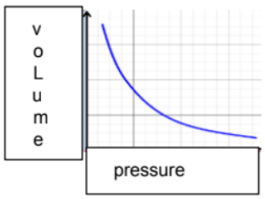
1. What did the person in the video do first with the balloon and bottle?
   - He attached the balloon to the mouth of an empty bottle.
2. Describe what happens to the balloon when it is placed in hot or cold water. Does it expand or shrink?
   - When the balloon is placed in hot water, it expands.
   - When the balloon is placed in cold water, it shrinks or deflates.
3. Why does the balloon expand when placed in hot water and shrink when placed in cold water?
   - The balloon expands in hot water because the air inside heats up and the molecules move faster.
   - The balloon shrinks in cold water because the air cools down and the molecules move slower.
4. Now, in the simulation shown to you, what happens to the size of the container (volume of gas) when the temperature is increased or decreased?
   - When the temperature is increased, the volume of gas increases, so the container expands.
   - When the temperature is decreased, the volume of gas decreases, so the container shrinks.
5. Based on the video and simulation, what is the relationship between the volume (size of the space) and the temperature of the gas? What happens to the volume when the temperature goes up or down?
   - Directly proportional

C. Abstraction (20 minutes)

The learners will analyze the content presented in the PowerPoint slides, listen to key explanations, and participate by answering questions in-between. They will also take turns reading and interpreting the information on the slides to support understanding. The following information is included in the slides:

A. Boyle's Law (Pressure – Volume Relationship)

- At constant temperature, the pressure of a gas is inversely proportional to its volume.
- A graph showing the relationship of pressure and temperature:



- The pressure of a gas is inversely proportional to its volume when temperature is constant.
- Example:

- Breathing:
    - **The Mechanism:** When your diaphragm muscle contracts and moves downward, it increases the **volume** of your chest cavity.
    - **Boyle's Law:** As volume increases, the air **pressure** inside your lungs decreases (becoming lower than the air outside).
    - **Result:** Air rushes into your lungs to equalize the pressure (Inhalation).

- The "Pop" of a Soda Bottle
    - **The Mechanism:** Soda is bottled under high pressure to force carbon dioxide gas to dissolve in the liquid. The gas is squeezed into a tiny space at the top of the bottle.
    - **Boyle's Law:** When you twist the cap, you break the seal, allowing the gas to escape into the atmosphere (a massive increase in volume).
    - **Result:** The pressure of the gas drops instantly. The gas expands rapidly, creating the "fizz" and foam as bubbles race to escape.

- Formula: P1V1 = P2V2

Where:
P1 = initial pressure
V1 = initial volume
P2 = final pressure
V2 = final volume

- **Example:**

At room temperature, a gas occupies 2.8 L at a pressure of 2.5 atm. What will be its final volume if the pressure is decreased to 1.3 atm?

Given: P1 = 2.5 atm | P2 = 1.3 atm

V1 = 2.8 L

Unknown: V2 = ?

Formula: P1V1 = P2V2

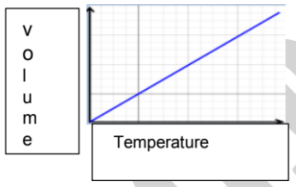
Solution: $V_2 = \frac{P_1 V_1}{P_2}$

$$V_2 = \frac{(2.5\cancel{atm})\,(2.5L)}{1.3\cancel{atm}}$$

Answer: 5.38 L

B. Charles' Law (Volume – Temperature Relationship)

- At constant pressure, the volume of a gas is directly proportional to its absolute temperature.



A graph showing the relationship between volume and temperature

- Example:

  - Hot Air Balloons
    - **The Mechanism:** A propane burner heats the air inside the giant balloon envelope.
    - **Charles's Law:** As the temperature of the air rises, its **volume** increases (it expands). Because the same amount of air now takes up more space, it becomes less dense than the cooler air outside.
    - **Result:** The balloon floats. To land, the pilot lets the air cool down, causing it to contract and become denser again.

  - The "Deflated" Basketball in Winter
    - **The Mechanism:** You leave a fully inflated basketball or football outside in the backyard overnight during cold weather.
    - **Charles's Law:** The cold temperature slows down the air particles inside the ball. They move less and take up less space, causing the **volume** of the gas to decrease.
    - **Result:** The ball looks "flat" or deflated in the morning, even though no air leaked out. (Once you bring it inside to warm up, it re-inflates)

- Formula:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Where:

V1 = initial volume

T1 = initial temperature (in Kelvin)

V2 = final volume

T2 = final temperature (in Kelvin)

- Example:

  A balloon is filled with 2.5 Liters of air inside an air-conditioned room at 25°C. The balloon is then taken outside on a hot summer day where the temperature is 40°C. What is the new volume of the balloon?

Given:

V1 = 2.5 L

T1 = 25°C = 298 K

T2 = 40°C = 313 K

Unknown:

V2 = ?

Formula:

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Derived Formula: $V_2 = \frac{V_1 \times T_2}{T_1}$

Solution: $V_2 = \frac{2.5\ \text{L} \times 313\ \text{K}}{298\ \text{K}}$

$$V_2 = \frac{782.5}{298}$$

$$V_2 \approx 2.63\ \text{L}$$

Answer: 2.63 L

D. Application (10 minutes)

Following the discussion, to test their understanding, the learners will solve problems related to Boyle's Law and Charle's Law. They will answer the following:

Boyle's Law:

1. A tank of oxygen has an initial volume of 300ml and a pressure of 8.2 atm. What is the final pressure in torr if the volume is doubly increased? (Answer: 3116 torr)
2. An air bubble released by a diver at the bottom of the ocean has a volume of 12 mL and a pressure of 4.0 atm. What is the final volume of the bubble in mL when it reaches the surface, where the pressure is 760 torr? (Answer: 48 ml)
3. A flexible container used for deep-sea sampling has an initial volume of 8.0 L and a pressure of 1.25 atm at the surface. What is the final pressure in torr if the container is compressed to one-fifth (1/5) of its original volume? (Answer: 4750 torr)
4. A weather balloon is filled with 600 mL of helium on the ground. As it rises into the atmosphere, the pressure drops to exactly one-third (1/3) of the ground pressure. What is the new volume of the balloon? (Answer: 1800 ml)

Charle's Law:

1. A scuba diver fills a small air pocket in his underwater helmet. At a temperature of 5 °C, the air expands to 1.8 L when warmed to 25 °C. If the initial volume of the air pocket is unknown, what was its volume at 5 °C? Assume the pressure remains constant. (Answer: V1 = 1.68 L)
2. A man blew up a balloon to 2 L at 300 K. When he left it under the hot sun, the temperature rose to 450 K. What is the new volume of the balloon? (Answer: V2 = 3 L)
3. Aiah finds her balloon in the freezer, and its volume is 2.5 L. She remembers that the balloon was originally 3.0 L at room temperature. The freezer's temperature is 250 K. What was the original temperature of the balloon before it went into the freezer? (Answer: T1 = 300 K)
4. A container has 5 L nitrogen gas at 25 °C. What will be its temperature if the volume increases by 10 L, keeping the pressure constant? (Answer: T2 = 596 K)

IV. Learning Assessment

Part I: Multiple Choice (1 point each item) 15 items

1. Who invented the barometer which was used to measure the atmospheric pressure?

A. Galileo Galilei
B. Isaac Newton
C. Evangelista Torricelli
D. Robert Boyle

2. In which century was the barometer invented?

A. 16th century
B. 17th century
C. 18th century
D. 19th century

3. How does the size of the space (volume) affect gas particle collisions?

A. Smaller space increases collisions, larger space decreases collisions.
B. Smaller space decreases collisions, larger space increases collisions.
C. Volume has no effect on collisions.
D. Particles stop moving if the space is too small.

4. Why does a balloon inflate when air is added?

A. The gas particles disappear inside the balloon.
B. Gas particles collide with the balloon walls, creating pressure that expands it.
C. Gas particles stop moving, allowing the balloon to expand.
D. The balloon creates gas particles from nowhere.

5. Boyle's Law states that at constant temperature, the pressure of a gas is _____ to its volume.

A. Directly proportional
B. Inversely proportional
C. Unaffected
D. Equal

6. Gas Pressure is defined as:

A. the force gas particles exert on the walls of their container.
B. the space occupied by the gas, which changes to fit the container.
C. the quantity of gas, usually measured in moles.
D. the average kinetic energy of gas particles

7. Which variable remains constant in Boyle's Law?

A. Pressure
B. Temperature
C. Volume
D. Energy

8. A flexible balloon contains gas at a certain temperature. If the absolute temperature (in Kelvin) of the gas is reduced to exactly 3/4 of its original value while the pressure remains constant, what will happen to the volume of the balloon according to Charles's Law?

A. The volume will increase to 4/3 of its original size.

B. The volume will remain the same.

C. The volume will be reduced to 3/4 of its original size.

D. The volume will drop to 1/4 of its original size.

9. A student uses a specialized container to expand a gas. If the volume of the gas is tripled (3x) while the mass remains constant, what happens to the density of the gas inside?

A. The density triples because the particles have more room to move.

B. The density remains the same because no gas particles were removed.

C. The density increases by nine times.

D. The density is reduced to 1/3 of its original value.

10. Why is it dangerous to throw an "empty" aerosol can into a fire, even if the valve is open?

A. The remaining gas particles move faster, increasing pressure inside.

B. The metal of the can reacts chemically with the air to create a vacuum.

C. The volume of the can increases until the metal reaches its melting point.

D. The gas inside the can turns into a solid, causing the can to collapse.

11. Imagine a weather balloon is launched into the upper atmosphere where the air is much thinner. If the balloon was fully inflated on the ground, what is the most likely outcome?

A. It will burst because the gas expands as the outside pressure drops.

B. It will shrivel up because it is colder in the sky.

C. It will sink back to the ground because the gas becomes heavier.

D. It will stay exactly the same size because it is sealed.

12. If you trap air in a syringe and pull the plunger out (increasing volume), the pressure drops. On a molecular level, why does the pressure decrease?

A. The particles grow smaller to fit the larger space.

B. The particles hit the walls less frequently because they move slower due to the increase of the volume of the container.

C. The particles hit the walls less frequently because they have to travel further between collisions.

D. The particles leak out through the plastic of the syringe.

13. A scientist is testing a new gas container. If she doubles the pressure applied to the gas while keeping the temperature steady, what should she expect to see happen to the volume?

A. The volume will double.

B. The volume will be reduced to one-half of its original size.

C. The volume will be reduced to one-fourth of its original size.

D. The volume will remain constant regardless of pressure.

14. During inhalation, the diaphragm contracts and moves downward, increasing the volume of the chest cavity. According to Boyle's Law, how does this facilitate breathing?

A. It increases the pressure in the lungs, forcing air out.

B. It decreases the pressure in the lungs, allowing outside air to rush in.

C. It increases the temperature of the lungs to attract oxygen.

D. It creates a vacuum that destroys carbon dioxide particles.

15. A gas sample undergoes a process where its pressure is reduced to 1/5 of its initial value. If the temperature is held constant, what is the mathematical expression for the new volume V_2 in terms of V_1?

A. $V_2 = 0.2V_1$

B. $V_2 = V_1 - 5$

C. $V_2 = 5V_1$

D. $V_1 = (⅕)V_2$

Part II: True or False (1 point each item) 5 items

1. The properties of gases can affect one another. True
2. Temperature describes the frequency of wall collisions of the particles. False
3. Kinetic Molecular Theory is a scientific model that explains the behavior of gases. True
4. If the temperature of a gas increases, its particles move slower. False
5. Gas particles in a container stop moving if the volume is too small. False

Part III: Problem Solving (5 points each - given: 1 point, asked/unknown: 1 point, formula: 1 point, solution with correct answer: 2 points) 3 items

1. A container has 5 L nitrogen gas at 25 °C. What will be its temperature if the volume increases by 10 L, keeping the pressure constant? (Answer: T2 = 596 K)
2. A gas is stored in a 500 ml piston at an initial pressure of 760 mmHg. The piston is compressed until the volume is exactly 1/4 of its original size. (4.0 atm)
3. A balloon has a volume of 2.5 L at a room temperature of 27° C. The balloon is placed in a freezer where the absolute temperature (in Kelvin) is reduced to exactly 4/5 of its original value. (2 L)

Part IV: Analysis (5 points each item) 2 items

1. During a PE game, someone accidentally steps on a ping-pong ball, leaving a large dent in it. The teacher doesn't throw it away. Instead, she drops the dented ball into a cup of boiling water.

After a few seconds, the dent pops out, and the ball is round again. How did the hot water fix the ball without anyone touching it? *(When the dented ping-pong ball is placed in boiling water, the air inside the ball heats up. The heat makes the air particles move faster and spread apart. As a result, the air expands and creates pressure from inside the ball. This pressure pushes outward on the dented surface, causing the ball to return to its round shape. No one needs to touch the ball because the expanding air does all the work.)*

2. According to the Kinetic Molecular Theory, if the temperature of a gas remains constant, the average speed of the particles does not change. If this is true, why does the pressure increase when the volume is decreased? (When volume decreases at a constant temperature, gas particles are crowded into a smaller space. Because the particles maintain the same average speed but have less distance to travel between walls, the frequency of collisions against the container surfaces increases. Since pressure is the result of these collisions, more frequent impacts result in higher pressure.)

V. Assignment

Directions: On a ½ crosswise paper, research Gay-Lussac's Law and write the following:

1. Who discovered the law and what it states
2. The importance of this law

Be ready to share your findings in the next class.